MUHAMMAD SAFWAN BIN MOHD AZMI

Aspiring Data Engineer | Cloud & Big Data

+6013-247-7532 · safwanazmi486@gmail.com · linkedin.com/in/safwanazmi

- https://safwan0486.github.io/portfolio/

PROFILE

Computer Science graduate specializing in Data Engineering, with hands-on industry experience in cloud-based ETL pipeline development, FinOps, big data processing, and real-time data engineering. Demonstrated ability to process datasets exceeding 22+ million records, develop data-driven solutions, and leverage technologies including Python, SQL, Apache Spark, Kafka, and AWS. Experienced in supporting cloud governance, data analytics, and enterprise project monitoring within the financial services sector. AWS Certified Cloud Practitioner and FinOps Certified Practitioner, bringing a strong analytical mindset, technical adaptability, and a passion for building scalable, efficient, and data-driven solutions. Seeking to apply cloud and data engineering skills within a leading telecommunications organization.

TECHNICAL SKILLS

Programming: Python, SQL, Linux, Java, C++, PHP, TypeScript, JavaScript

Data Engineering & Big Data: Apache Spark, Apache Airflow, Apache Kafka, Pyspark, Pandas,Dask ,Polars

Databases & Search: MySQL, PostgreSQL, MongoDB Atlas, Elasticsearch

Cloud & Devops: AWS (EC2, GLUE,IAM) ,Azure, Docker , Kubernates

Data Visualization: Power BI, Tableau, Kibana

Web Scrapping: Selenium, BeautifulSoup, Scrapy

Tools: Git, Visual Studio Code, SAP, Sharepoint, Figma

EXPERIENCE

Industrial Attachment - Bank Pertanian Malaysia Berhad (Agrobank) **Sep 2025 - Jul 2026**

- Developed a FinOps ETL data pipeline to monitor cloud spending, supporting cost-optimization decisions across cloud projects
- Configured cloud environments and conducted cloud readiness and risk assessments for financial services workloads
- Performed data cleaning and preprocessing datasets to build interactive dashboards used for performance monitoring and decision support
- Built a project monitoring dashboard using SharePoint, integrated with Power BI to track progress across 20+ projects
- Assisted in delivering the IESP (Innovation Emerging Service Process) flow for cloud projects, including cloud strategy and cloud framework submissions to Bank Negara Malaysia (BNM)
- Conducted third-party cloud governance gap analysis aligned with RMiT (Bank Negara Malaysia) guidelines
- Documented end-to-end cloud migration processes, strengthening compliance tracking and audit readiness

Program Akar - AWS re/Start (Paynet) **Feb 2025 - Aug 2025**

- Completed hands-on training in core AWS services (EC2, S3, IAM, Lambda), system operations, and cloud security
- Applied cloud fundamentals to real-world projects, including exposure to digital payment systems

PROJECTS

Real-Time Sentiment Analysis System **Oct 2025**

- Built a real-time data pipeline using Apache Kafka, Spark, Elasticsearch, and Kibana to analyze YouTube comments
- Developed a TF-IDF + Logistic Regression sentiment classifier achieving 87% accuracy
- Deployed the system with Docker and visualized sentiment trends in real time

High-Performance Web Crawler (100K+ Records) **Oct 2025**

- Built a large-scale web crawler using Selenium, BeautifulSoup, and Scrapy, collecting 100,000+ records
- Processed data with PySpark and Dask, storing results in MongoDB Atlas

Big Data Processing & Performance Benchmarking **Aug 2025**

- Processed a 2.41GB dataset (22M+ records) to evaluate big data optimization techniques
- Benchmarked Pandas, Dask, and Polars; identified Polars as fastest with the lowest memory usage

Smart Eco Vision System - AI-Based Agricultural Monitoring **Feb 2024**

- Built an AI-powered agricultural monitoring system for plant health assessment using machine learning and IoT sensors
- Integrated image recognition for detecting plant abnormalities and real-time soil monitoring

Additional projects: Exita Manufacturing Inventory & Procurement System (SAP ERP, Mar 2025) · AKMaju Construction & Advertisement Management System (Jan 2024)

EDUCATION

Bachelor of Computer Science (Data Engineering) with Honours
Universiti Teknologi Malaysia (UTM), Skudai, Johor Bahru - CGPA 3.78 / 4.00

- Dean's List Award recipient

Foundation in Engineering
Universiti Teknologi MARA (UiTM), Dengkil, Selangor - CGPA 4.00, MUET Band 4

Sijil Pelajaran Malaysia (SPM)
Tuanku Jaafar Technical Institute (Electrical and Electronic Engineering)

EXTRACURRICULAR / VOLUNTEER ACTIVITIES

CyberX Competitor - Capture The Flag (CTF) Team **Dec 2024**
Universiti Teknologi Malaysia (UTM)

- Represented UTM in CyberX CTF 2024, competing in Web Exploitation, Cryptography, Forensics, and Reverse Engineering
- Applied penetration testing and incident investigation techniques to solve cybersecurity challenges

Service-Learning Futsal Program - Coach & Organizer **Feb 2024**
In collaboration with Kolej Matrikulasi Johor

- Conducted futsal training and strategy sessions for students
- Organized a futsal tournament, developing leadership and event-planning skills

Opera 23 - Main Committee Member **Dec 2023**

- Managed participant registration and merit allocation for event recognition
- Assisted with event logistics and operations, ensuring smooth execution

ADDITIONAL INFORMATION

Technical Skills: C++, Java, Python, PHP, HTML/CSS, SQL, Tableau, Power BI, RStudio, Figma, SAP, Visual Studio Code
Languages: Bahasa Melayu (Native), English (Professional Working Proficiency), Arabic (Elementary)
Certifications:

- AWS Certified Cloud Practitioner
- AWS re/Start Graduate
- AWS Academy Cloud Foundations
- AWS Academy Cloud Developing
- AWS Academy Cloud Data Pipeline Builder
- Generative AI Essentials on AWS
- FinOps Certified Practitioner
- Alteryx Designer Core Certification
- Alteryx Foundational Micro-Credential, issued by Alteryx
- ICDL Document
- ICDL Spreadsheet
- ICDL Teamwork